Fiscal Year:	FY 2010	**Task Last Updated:**	FY 06/07/2010
PI Name:	McKay, David Ph.D.		
Project Title:	Geology, Geochemistry and Lithology Science Support Activities		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Environmental health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	PI is deceased (Ed., Feb. 2013)		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/02/2006	**End Date:**	09/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Changed end date back to 9/30/2012 since this portion of project has been completed (Ed. 1/15/14) NOTE: End date changed to 9/30/2013 per John James (Ed., 8/24/2012) NOTE: End date changed to 9/30/2011 per JSC HRP info--MTL dtd 10/7/11 (Ed., 10/28/11) NOTE: Start/end dates changed to 10/2/2006-12/31/2010 (previously 4/30/2006-1/31/2011) per B. Woolford/JSC via S. Steinberg-Wright/JSC (9/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Taylor, Lawrence (University of Tennessee –Knoxville)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Lunar Airborne Dust Toxicity Assessment Group (LADTAG) has been assigned the task of evaluating dust hazards to humans, and recommending exposure standards for lunar dust suspended in breathing air. Over the 40 years that have transpired since the last Apollo crew left the lunar surface, much has been learned about the toxicity of small particles. Based on experience with industrial minerals such as sandblasting quartz, we know that even if a dust is not hazardous due to a three-day exposure, it may nonetheless be quite toxic when the exposure covers a longer duration. The Lunar Airborne Dust Toxicity Assessment Group (LADTAG) was chartered by NASA to set an exposure standard for 6 months of episodic exposure to lunar dust. In order to accomplish this, it is necessary to learn how lunar dust toxicity compares to both toxic and non-toxic terrestrial dusts. Dust exposure standards, similar to those established for quartz and asbestos on Earth, will determine the design, mass, power, and cost of airlocks, ports, and dust control systems that are incorporated into lunar habitats and shirtsleeve rovers. The Lunar Airborne Dust Toxicity Advisory Group (LADTAG) Geology team, led by JSC senior scientist David S. McKay, encompasses expertise in geology, geochemistry, and geophysics, with a combined total of over 100 years of lunar research. This team provides three major research thrusts for the LADTAG research effort: (1) Evaluation and Recommendations; (2) Isolating the respirable fraction of lunar dust for inhalation toxicology studies, and (3) Lunar soil simulant development. (1) Evaluation and Recommendations: Lunar dust in situ is subjected to processes that likely cause grain surfaces to become chemically reactive, potentially increasing the toxicity of the dust. Moreover, lunar dust contains metallic iron in spheroids of 30 nanometers and smaller diameter. The effect of this nanophase iron on human tissue and cells is yet to be determined; such material does not occur on Earth and it is thus unknown whether humans have any specific evolutionary ability to deal with it. The detailed characteristics of sub-micrometer lunar dust are only poorly known, and this is the size range of particles that are of greatest concern for our toxicology studies. In addition to being potentially hazardous to humans, lunar dust is detrimental to equipment. Electrostatic charging of the dust due to UV irradiation and solar-wind-proton implantation must be characterized in order to design methods of mitigating these effects. Our team will provide high-fidelity simulants by continuing our research in the effects of radiation, proton bombardment, and triboelectric charging. One of the most unusual features of actual lunar dust is the presence of minute iron spheres, up to 30 nanometers in diameter, which often cover the surface of the grain. Our team has simulated these iron spheres in a silicate matrix for use in toxicology studies. Our activities in support of Evaluations and Recommendations include basic research on chemical activation, passivation, size and shape analysis, and simulating the unique qualities of the surfaces of lunar dust grains as it will be found on the lunar surface. In addition to providing knowledge about past research on lunar dust, our team will be conducting new experiments to address unknowns that have not previously been considered. For example, we plan dissolution studies, both with dust simulant and with actual lunar dust, to learn more about how dust particles behave when ingested. This provides critical background information for toxicology studies. (2) Isolating the respirable fraction of lunar dust for inhalation toxicology studies: Inhalation toxicology studies using lunar dust will be used to determine the allowable exposure limits. These experiments must be performed in a standardized manner so that the results obtained can be realistically compared to the results of experiments using other substances (such as silica dust). Consequently it is necessary to separate the respirable particles from a lunar sample, because the typical laboratory equipment for inhalation studies depends on having particles in the proper size range at the start of the experiment. Our efforts focus on dust in the respirable range and include the particle size distribution, mineralogical and morphological habits of the dust in lunar-habitat conditions, dust reactivity and passivation in applicable conditions, and direct cell interactions with nano-scale-iron-rich minerals and glasses. This allows LADTAG to define risks and establish appropriate standards for manned lunar missions. These analyses allow us to prepare lunar soils with the targeted size distributions and properties for use by the toxicology groups for inhalation and cell reaction studies. An additional by-product is the development of improved techniques for separating and handling nano-scale particles for Scanning Eletron Microscope (SEM), Transmission Electron Microscope (TEM), and elemental analysis. These requirements have inspired us to develop a method of separating the respirable fraction of lunar dust from a bulk soil sample, without using sieves, which allows maximum recovery of the available dust in the sample. The total amount of lunar sample available for this study is less than 300 grams. The developed method uses pneumatic transport with ultrapure nitrogen as the carrier gas; thus the lunar material is not exposed to water or other liquids. Our separation method incorporates a small fluidized bed as well as gas impactor size separation steps and a cyclonic separator. Analysis of the separated material with a Microtrac™ laser diffraction system has been used to validate the process. (3) Lunar soil simulant development: The geology team is developing new methods to simulate the chemically active surface of dust grains. Specifically, the areas of concern for LADTAG are the physical and chemical properties of the finest fraction (less than 3 μm) of lunar soil (“dust”), and the interaction of lunar dust with biological as well as mechanical systems. Before the hazards of lunar dust can be managed, they must be understood. The path to understanding the hazards includes both experimental and analytical work. The experimental work cannot be performed on actual lunar samples without first validating the proposed techniques using an appropriate simulant material. By developing an appropriate dust simulant, the Geology Team provides the basis for all other areas of research within LADTAG.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	The separation method can be used for other purposes, and can be modified to handle larger or smaller quantities of material. It is particularly useful for isolating statistically representative particles for Transmission Electron Microscopy (TEM) studies. The system could also be modified to collect a different size fraction. Follow-on applications for this system include actual lunar surface use, perhaps as part of a pneumatic mining and soil transport system on the moon as the central core for lunar ISRU hydrogen extraction, and industrial waste clean-up (Cuero and McKay 2007).

Task Progress:	A. Dust Separation We have successfully separated a total of 2.45 grams of respirable dust from 200 gm lunar sample 14003, 96, which was acquired at the Apollo 14 landing site. This yield exceeds our minimum initial estimate. This fine-grained dust will be primarily used for toxicology studies and will also be characterized for nanoparticle size distribution, bulk chemistry, grain chemistry, mineralogy, optical reflectance, nanophase iron content, and other critical properties. Our method for extracting respirable dust (<2.0 µm) from Apollo lunar soils meets stringent requirements: the soil must be kept dry, the soil should only be exposed only to pure nitrogen gas, and the soil must be conserved and recovered to the maximum extent possible. Because dry sieving is not very effective at releasing the smallest dust particles that are adhered to larger grains, we selected a series of components that would be more effective than traditional sieving. One of the components is a prototype fluidized bed which was developed at JSC for use as an in-situ resource utilization (ISRU) demonstration unit. Modifications to the original component include the addition of a shaped filter holder at the bottom, which prevents material from being caught in the corner space at the bottom edge. Previous experience had shown us that these "dead" spots would eventually lead to channeling, in which fluidization stops and the gas forms a channel to the top of the bed (Cooper 2008). The modification has resulted in continuous fluidization behavior for a minimum of four hours. Our present pneumatic size sorting system starts with a small fluidized bed using only inert gas. Other components include a flat plate impactor and a small cyclone separator. Separation of lunar soil was begun on 24 February 2010. Approximately 200 gm of 14301, 96 was put into the first stage (fluidized bed) in the nitrogen cabinet. After allowing the process to run for four hours, the dust on the membrane filter was scraped or brushed off and placed inside stainless steel sample containers which have Teflon snap-off lids. Later, each sample container was double-bagged while still in ultrapure dry nitrogen, using Teflon bags. B. Grinding We have developed a method for grinding coarser lunar soil to produce sufficient respirable soil for animal toxicity testing while preserving the freshly exposed grain surfaces in a pristine state. Using this technique, we have produced 55 grams of material that has a median diameter of approximately 2 micrometers. In conjunction with this development, we have produced a second soil-separation system that is useful for particles that are already quite small. The new separation system allows us to extract grains of 1.8 micrometers or smaller from a sample with a median particle diameter of 3.5 micrometers. This separation is most useful for rodent studies, because the 1.8 micrometer diameter represents the size of particle that is respirable by a mouse. We evaluated a number of methods for grinding lunar soil and chose a jet-mill, in which the soil is entrapped in intersecting gas streams of pure, dry nitrogen. This action causes the grains to be impacted and ground primarily by collisions with other grains rather than by interactions with a grinding medium such as alumina, tungsten carbide, or agate. Contamination by non-lunar grinding media is minimized in this process.
Bibliography Type:	Description: (Last Updated: 06/08/2010)
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Articles in Peer-reviewed Journals	Wallace WT, Phillips CJ, Jeevarajan AS, Chen B, Taylor LA. " Nanophase iron-enhanced chemical reactivity of ground lunar soil." Earth and Planetary Science Letters, In Press, Corrected Proof, Available online 23 May 2010. http://dx.doi.org/10.1016/j.epsl.2010.04.042 , Jul-2010
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Patents	US Patent Office, Patent Number 7,309,437 B2. Patent, December 2007. Dec-2007 Cuero R, McKay DS. "Compositions and Methods for Removal of Toxic Metals and Radionuclides."